



On-policy Distillation of Language Models: Learning from Self-Generated Mistakes

大语言模型在线蒸馏：从错误中自学习

ON-POLICY DISTILLATION OF LANGUAGE MODELS:
LEARNING FROM SELF-GENERATED MISTAKES

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什么是同策略？



同/离策略最早出现在强化学习领域，同策略值的是在环境中采样的策略与训练时Actor的策略是同一策略；离策略则表示二者不是同一策略。在强化学习语境下，同策略通常指的是“同步更新”。

因此，将这个概念加以推广：**同策略训练**从学生模型本身采样展开轨迹，并为其分配一定的奖励；而**离策略训练**依赖于某个外部来源的目标输出，学习体通过模仿这些输出来学习。

常见的离策略训练如SFT，它依赖于固定的专家轨迹；同策略训练如我们即将介绍的同策略蒸馏，它通过学生模型采集轨迹，再由教师模型为每个词元评分。

动机

- 散度选择：

- 传统知识蒸馏选择前向KL散度，公式为： $D_{KL}(P||Q) = \int P(x) \log \frac{P(x)}{Q(x)} dx$ ，P为教师模型，Q为学生模型。
当 $P(x) > 0$ ，只要 $Q(x)$ 趋近于0，损失就会趋于无穷大，因此即使是教师模型分配概率很低的词元上，学生模型也被迫要分配概率。当学生模型远小于教师时，容易导致幻觉和低质量生成。

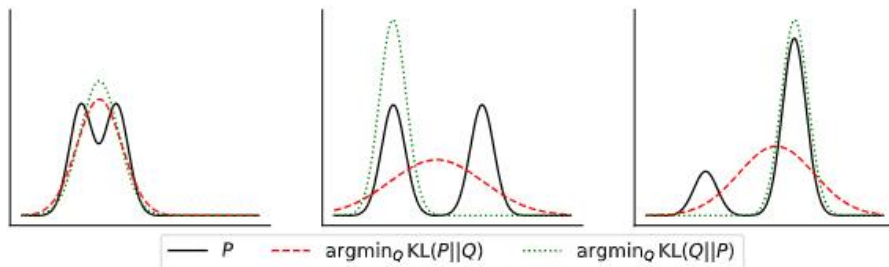


Figure A.16: **Mode-seeking vs Model-covering KL with capacity mismatch.** We show the learned distribution Q_θ when minimizing the forward and reverse KL w.r.t Q between a mixture distribution P and a unimodal Gaussian Q_θ . Reverse KL is mode-seeking as it forces Q_θ to be zero where P is zero and hence makes it concentrate on one of the modes (last plot). However, forward KL is mode-covering as it ensures that there is some mass under Q_θ wherever there is some mass under P . See [Le \(2017\)](#) to replicate this plot.

动机



- 分布不匹配：
 - 传统知识蒸馏均依赖固定输出序列（真实标签或教师生成序列）训练，与推理阶段学生自回归生成的序列分布不一致，引发暴露偏差与错误级联，严重降低生成质量。数学上讲，对于监督知识蒸馏，训练时损失函数为 $\frac{1}{L} \sum_{n=1}^L D_{KL}(p_t(\cdot | x, y_{<n}) || p_s(\cdot | x, y_{<n}))$ ，其中 $y_{<n}$ 来自教师分布 p_t ；而在推理阶段，来自学生分布 p_s ，从而造成误差。
- 已在其他领域应用：
 - 同策略模仿学习方法通过学生模型策略迭代采集序列，为这些序列获取专家标签，再在该数据集上重新训练学生模型。尽管这类方法在机器人学与深度强化学习领域已得到广泛应用，但通常并未用于蒸馏自回归模型。



Algorithm 1 Generalized Knowledge Distillation (GKD)

```

1: Given: Teacher model  $p_T$ , Student Model  $p_S^\theta$ , Dataset  $(X, Y)$  containing (input, output) pairs
2: Hyperparameters: Student data fraction  $\lambda \in [0, 1]$ , Divergence  $\mathcal{D}$ , Learning rate  $\eta$ 
3: for each step  $k = 1, \dots, K$  do
4:   Generate a random value  $u \sim \text{Uniform}(0, 1)$ 
5:   if  $u \leq \lambda$  then
6:     Sample inputs  $x$  from  $X$  and generate outputs  $y \sim p_S^\theta(\cdot|x)$  to obtain  $B = \{(x_b, y_b)\}_{b=1}^B$ 
7:   else
8:     Sample batch of inputs and outputs from  $(X, Y)$  to obtain  $B = \{(x_b, y_b)\}_{b=1}^B$ .
9:   end if
10:  Update  $\theta$  to minimize  $L_{\text{GKD}}$ :  $\theta \leftarrow \theta - \eta \frac{1}{B} \sum_{(x,y) \in B} \nabla_\theta \mathcal{D}(p_T \| p_S^\theta)(y|x)$ 
11: end for

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$$L_{\text{GKD}}(\theta) := (1 - \lambda) \mathbb{E}_{(x,y) \sim (X,Y)} [\mathcal{D}(p_T \| p_S^\theta)(y|x)] + \lambda \mathbb{E}_{x \sim X} \left[\mathbb{E}_{y \sim p_S(\cdot|x)} [\mathcal{D}(p_T \| p_S^\theta)(y|x)] \right],$$

此外，仅依靠教师模型进行知识蒸馏，本质上只是对我们核心优化目标（换句话说就是“真实目标”）的一种间接逼近。对此，我们可以直接通过RL对目标进行优化。
若希望在贴近教师模型策略的前提下，针对标量奖励 r 优化学生模型策略，便可得到如下带正则化的强化学习微调目标：

$$\mathbb{E}_{x \sim X} \left[\underbrace{(1 - \alpha) \mathbb{E}_{y \sim p_S^\theta(\cdot|x)} [r(y)]}_{\text{RL objective}} - \underbrace{\alpha \mathbb{E}_{y \sim p_S(\cdot|x)} [\mathcal{D}(p_T \| p_S^\theta)(y|x)]}_{\text{Generalized On-Policy Distillation}} \right],$$

从中其实也可以看出，OPD可以作为强化学习的一种替代，相当于用负的反向KL替代RL的奖励函数。

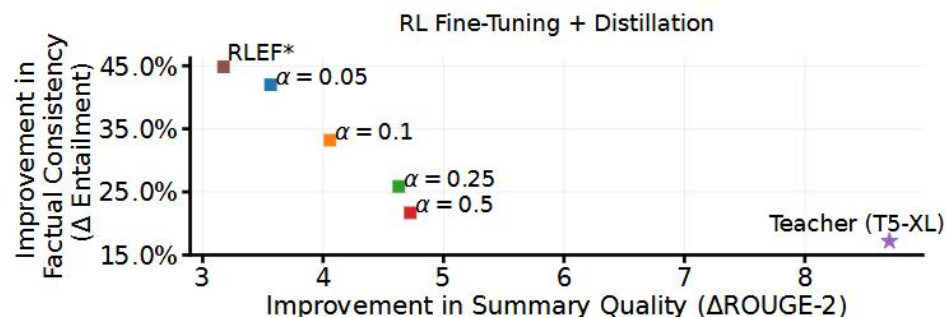


Figure 5: **RLAIF + On-policy GKD**. We show the trade-off between reward maximization and summarization performance on XSum. We report improvements relative to the original T5-base student. Following Roit et al. (2023), we use the textual entailment score from a T5-XXL NLI classifier as the reward. α controls the strength of the on-policy GKD loss with JSD (0.9). As α increases, ROUGE-2 increases while improvement in factual consistency decreases. For comparison, we show the relative performance of the 12× larger T5-XL teacher. RLEF* corresponds to RLAIF method from Roit et al. (2023), where the student is regularized towards the original student model itself instead of the teacher. On-policy GKD + RL achieves higher ROUGE-2 compared to RLEF* while generating more factually consistent summaries compared to the teacher.

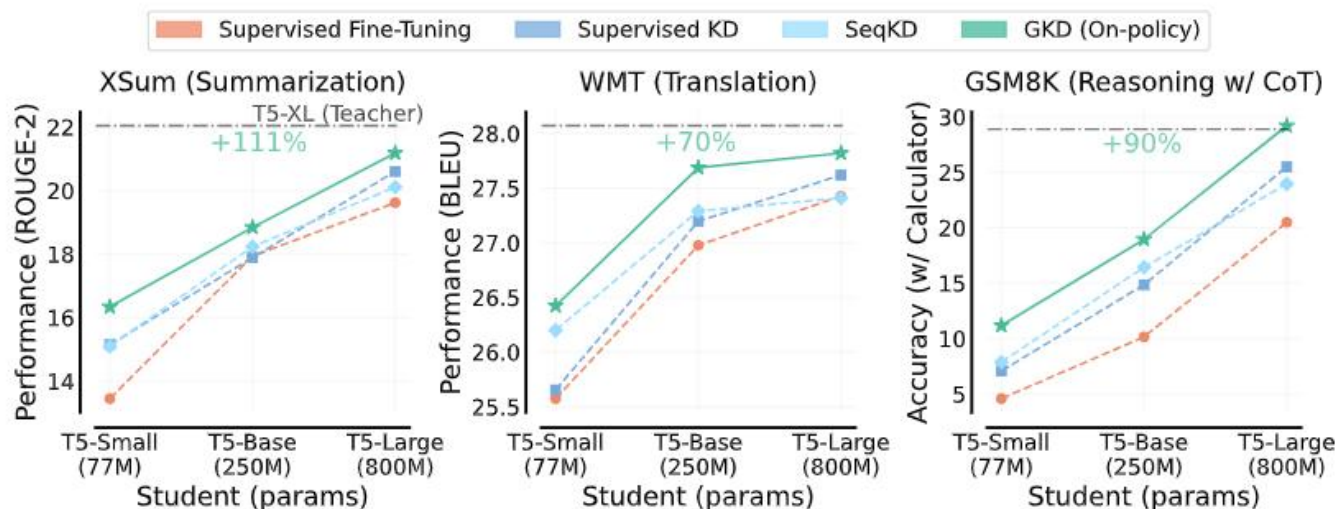


Figure 1: **Comparing GKD with KD approaches** across different student model sizes. We use the T5 models (Raffel et al., 2020) trained with supervised FT as students. We use supervised FT T5-XL (~3B params) as the teacher, whose performance is indicated by the horizontal line. Supervised KD and FT use ground-truth output sequences for training while SeqKD trains on output sequences generated by the teacher. On-policy GKD trains on output sequences sampled from the student. For GKD, we use JSD (0.1) on WMT and forward KL on other tasks. For evaluation, we use greedy sampling for XSum and GSM8K and beam search for WMT.

这张图主要对比了GKD和其他知识蒸馏方法的性能，验证了方法的有效性。值得注意的是针对不同数据集，本文选取了不同KL散度，也应证了KL的选取是任务依赖的。

03 实验结果

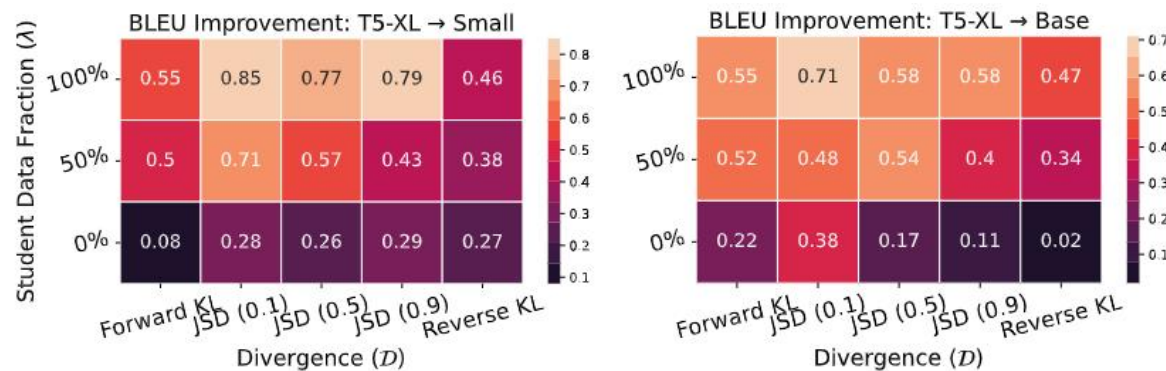


Figure 6: **Varying student data fraction and divergence in GKD on WMT en $\rightarrow$ de.** For evaluation, we use beam search and report the improvement in BLEU score of distilled student relative to the original student. Results are averaged across three seeds. We observe that using only student-generated output samples outperform other GKD variants. We use the T5-XL (~ 3 B params) supervised fine-tuned on WMT as the teacher, which obtains a BLEU score of 28. (Left) We use T5-small (77M params) as the student, which obtain a BLEU score of 25.58. (Right) Student corresponds to T5-base (250M params) with a BLEU score of 26.98.

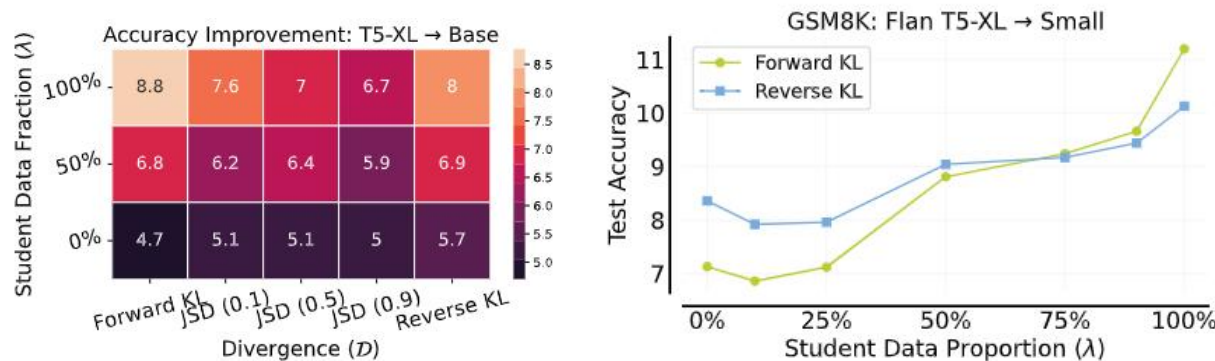


Figure 7: **Ablating GKD on GSM8K.** We distill fine-tuned T5-XL to T5-Base, which obtain an accuracy of 27.9 and 10.16 with greedy sampling.

Figure 8: **Varying on-policy data on GSM8K.** As we increase fraction of student-generated data beyond 25%, performance typically improves.

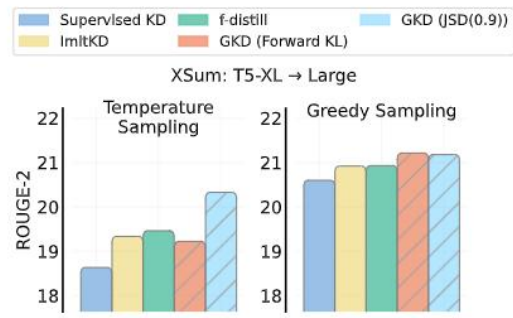


Figure 2: Comparing GKD to baselines on distillation from T5-XL to T5-large on XSum. On-policy GKD variants generally outperform baselines.

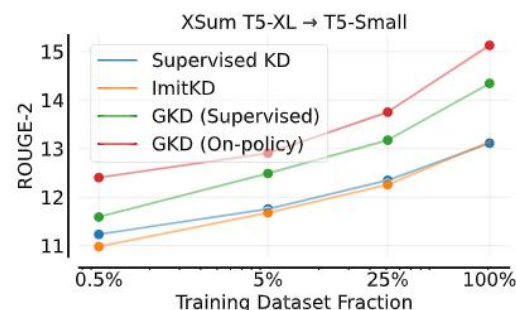


Figure 3: Scaling training data. We evaluate distilled T5-small using temperature sampling ($\gamma = 1$). GKD is more data efficient than baselines.

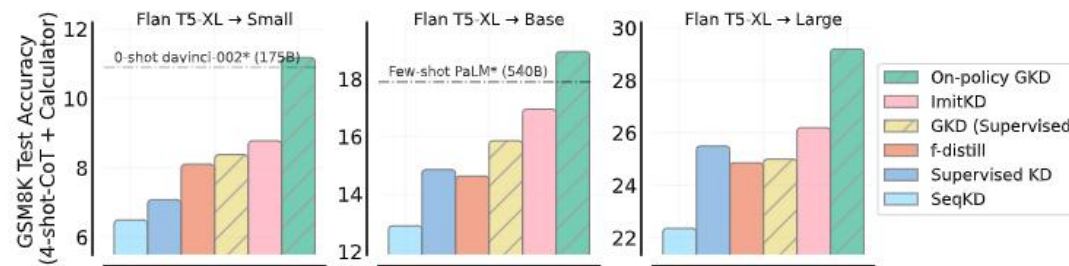


Figure 9: Distillation on GSM8K with few-shot CoT prompting. On-policy GKD substantially outperform other approaches. As a reference, we provide GPT-3 davinci-002 results as well as PaLM (540B) results (without a calculator). We use forward KL and reverse KL respectively for on-policy and supervised GKD.

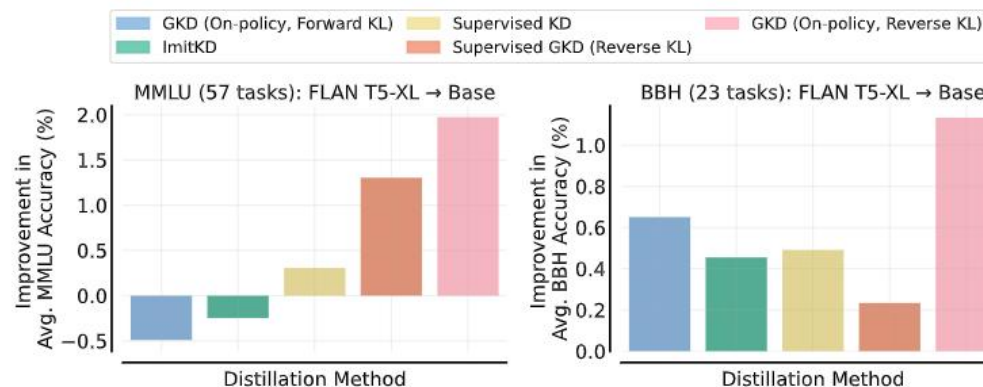


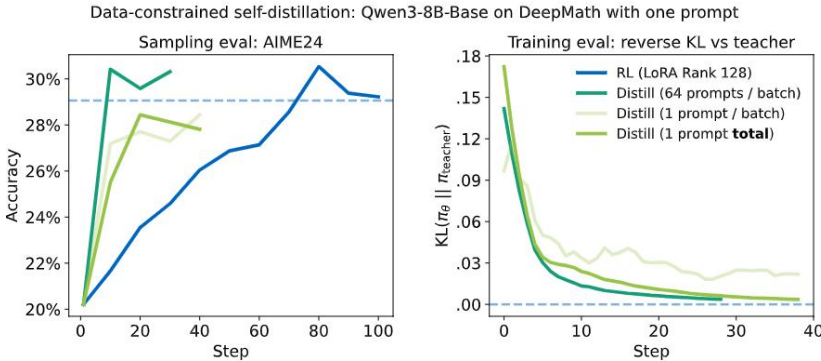
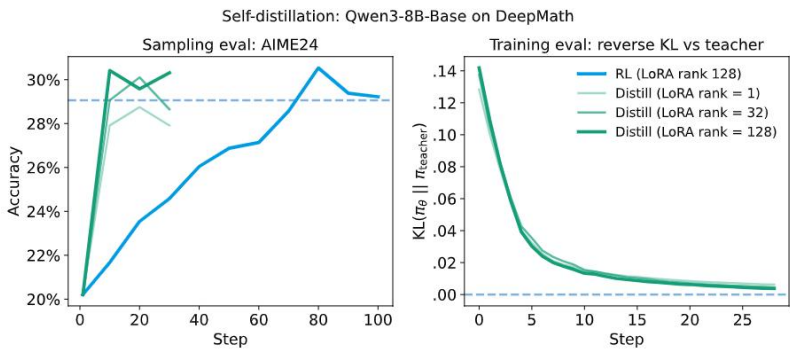
Figure 10: Task-agnostic Distillation on FLAN (Chung et al., 2022). On-policy GKD with reverse KL outperforms other approaches. The evaluation metric on both the MMLU and BBH benchmark suites is few-shot prompted accuracy (exact match), where we take an unweighted average over all tasks. These evaluation benchmarks are held-out (not included in the distillation data). Here, we do not run SeqKD due to its computational inefficiency for generating data from the teacher during training. The teacher FLAN T5-XL achieves an accuracy of 52.4% on MMLU and 41% on BBH, while the student T5-large model obtains an accuracy of 35.6% on MMLU and 31.25% on BBH.

与强化学习对比:

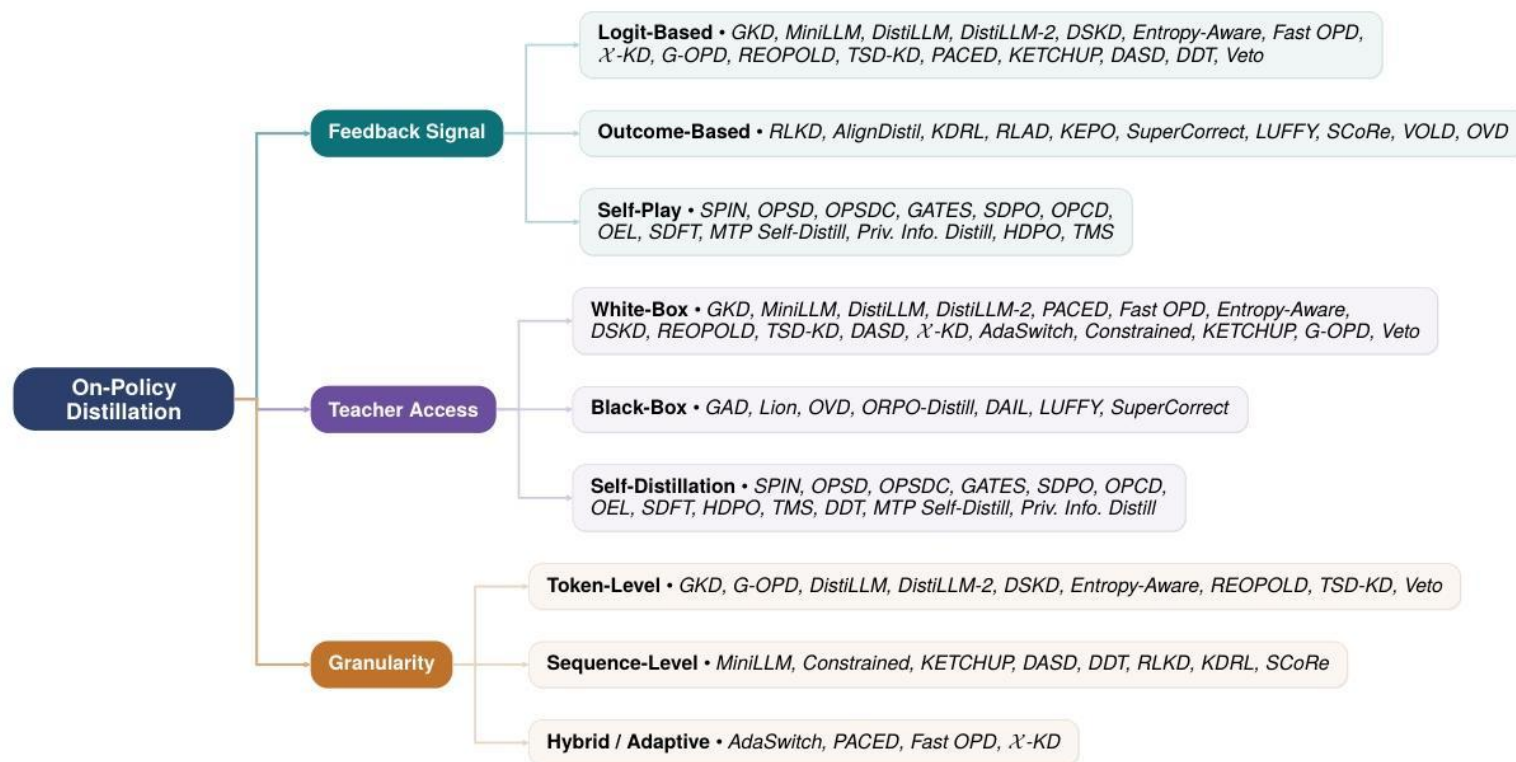
Method	Sampling	Reward signal
■ Supervised finetuning	off-policy	<u>dense</u>
■ Reinforcement learning	<u>on-policy</u>	sparse
■ On-policy distillation	<u>on-policy</u>	<u>dense</u>

OPD相较于RL，由于OPD的奖励是逐词元的，它的奖励更加稠密，因此带来两大优势：

- **计算开销低**：对于每条轨迹，RL的策略只有O(1)的监督信号反馈，但OPD有O(N)密度的反馈。蒸馏算法达到教师模型性能水平的速度大约快7-10倍。
- **数据利用率高**：在强化学习中，对同一提示进行多次训练往往会导致模型简单地记忆最终答案，尤其是在模型规模较大时。相比之下，基于策略的蒸馏通过最小化反向KL分布来学习逼近教师的完整分布，而不是记忆单个答案。这使我们能够从同一个提示中训练多个样本。



上述作为引入，可以看到OPD具有很多优越性，很具有吸引力。但其实OPD还有一些待解决的问题，比如奖励信号粒度的选择、教师模型是黑盒还是白盒、教师流向学生的信息内容、KL散度的选择等等。目前OPD的发展路径主要体现在以下三个方面：





Scaling Reasoning Efficiently via Relaxed On-Policy Distillation

Scaling Reasoning Efficiently via Relaxed On-Policy Distillation

Jongwoo Ko Sara Abdali Young Jin Kim Tianyi Chen Pashmina Cameron



Microsoft

标准RL目标函数:

$$\mathcal{J}_{\text{RL}}(\theta) = \mathbb{E}_{q \sim \mathcal{Q}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{l=1}^{|o_i|} \rho_{i,l}(\theta) \hat{A}_{i,l} \right], \quad (1) \quad \text{where } \rho_{i,l}(\theta) = \frac{\pi_{\theta}(o_{i,l}|q, o_{i,<l})}{\pi_{\theta_{\text{old}}}(o_{i,l}|q, o_{i,<l})}$$

OPD目标函数:

$$\mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_T) = -\mathbb{E}_{q \sim \mathcal{Q}, o \sim \pi_{\theta}(\cdot|q)} \left[\log \frac{\pi_T(o|q)}{\pi_{\theta}(o|q)} \right]. \quad (2)$$

OPD相当于用负的反向KL
替代RL的奖励函数:

$$\mathcal{J}_{\text{RKL}}(\theta) = \mathbb{E}_{q \sim \mathcal{Q}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{l=1}^{|o_i|} \rho_{i,l}(\theta) R_{i,l}(\theta) \right], \quad (3) \quad \text{where } R_{i,l}(\theta) = \log \frac{\pi_T(o_{i,l}|q, o_{i,<l})}{\pi_{\theta}(o_{i,l}|q, o_{i,<l})}$$

然而, $R_{i,t}(\theta)$ 依赖于参数 θ , 我们仍显式地采用固定奖励视角来稳定优化过程。具体而言, 我们引入梯度截断算子, 将 $R_{i,t}(\theta)$ 视作一个恒定的外部信号。用 $\mathcal{J}_{\text{RKL}}^{(\text{sg})}$ 表示这一加入梯度截断后的优化目标, 则有如下性质成立:

$$\nabla_{\theta} \mathcal{J}_{\text{RKL}}(\theta) = \nabla_{\theta} \mathcal{J}_{\text{RKL}}^{(\text{sg})}(\theta) = \mathbb{E}_{q \sim \mathcal{Q}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{l=1}^{|o_i|} \rho_{i,l}(\theta) R_{i,l}(\theta) \nabla \log \pi_{\theta}(o_l|q, o_{<l}) \right],$$

$$\mathcal{J}_{\text{RKL}}^{(\text{sg})}(\theta) = \mathbb{E}_{\substack{q \sim \mathcal{Q} \\ \{o_i\} \sim \pi_{\theta_{\text{old}}}}} \left[\frac{1}{\sum |o_i|} \sum_{i,l} \rho_{i,l}(\theta) \cdot \text{sg}(R_{i,l}(\theta)) \right]$$
$$\nabla_{\theta} \text{sg}(x) = 0, \quad \text{前向值仍为 } x$$

We derive the gradient of RKL. When adopting similar techniques as PPO or GRPO, the RKL objective and its gradient are:

$$\mathcal{J}_{\text{RKL}}(\theta) = \mathbb{E}_{q \sim Q, o \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \rho_l(\theta) R_l(\theta) \right], \quad (11)$$

where $\rho_l(\theta) = \frac{\pi_{\theta}(o_l|q, o_{<l})}{\pi_{\theta_{\text{old}}}(o_l|q, o_{<l})}$ and $R_l(\theta) = \log \frac{\pi_T(o_l|q, o_{<l})}{\pi_{\theta}(o_l|q, o_{<l})}$. Then following holds,

$$\nabla_{\theta} \mathcal{J}_{\text{RKL}}(\theta) = \mathbb{E}_{q \sim Q, o \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \rho_l(\theta) \nabla_{\theta} R_l(\theta) + R_l(\theta) \nabla_{\theta} \rho_l(\theta) \right], \quad (12)$$

$$= \mathbb{E}_{q \sim Q, o \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \rho_l(\theta) R_l(\theta) \nabla_{\theta} \log \pi_{\theta}(o_l|q, o_{<l}) \right] - \mathbb{E}_{q \sim Q, o \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \frac{\nabla_{\theta} \pi_{\theta}(o_l|q, o_{<l})}{\pi_{\theta_{\text{old}}}(o_l|q, o_{<l})} \right], \quad (13)$$

$$= \mathbb{E}_{q \sim Q, o \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \rho_l(\theta) R_l(\theta) \nabla_{\theta} \log \pi_{\theta}(o_l|q, o_{<l}) \right] - \nabla_{\theta} \mathbb{E}_{q \sim Q, o \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \frac{\pi_{\theta}(o_l|q, o_{<l})}{\pi_{\theta_{\text{old}}}(o_l|q, o_{<l})} \right], \quad (14)$$

$$= \mathbb{E}_{q \sim Q, o \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \rho_l(\theta) R_l(\theta) \nabla_{\theta} \log \pi_{\theta}(o_l|q, o_{<l}) \right] \quad (15)$$

$$- \nabla_{\theta} \mathbb{E}_{q \sim Q, o \sim \pi_{\theta}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \frac{\pi_{\theta}(o_l|q, o_{<l})}{\pi_{\theta_{\text{old}}}(o_l|q, o_{<l})} \cdot \frac{\pi_{\theta_{\text{old}}}(o_l|q, o_{<l})}{\pi_{\theta}(o_l|q, o_{<l})} \right], \quad (16)$$

$$= \mathbb{E}_{q \sim Q, o \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \rho_l(\theta) R_l(\theta) \nabla_{\theta} \log \pi_{\theta}(o_l|q, o_{<l}) \right] - \nabla_{\theta} \mathbb{E}_{q \sim Q, o \sim \pi_{\theta}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} 1 \right], \quad (17)$$

$$= \mathbb{E}_{q \sim Q, o \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{|o|} \sum_{l=1}^{|o|} \rho_l(\theta) R_l(\theta) \nabla_{\theta} \log \pi_{\theta}(o_l|q, o_{<l}) \right] \quad (18)$$

$$(19)$$



$$\nabla_{\theta} \log \pi_{\theta} = \frac{\nabla_{\theta} \pi_{\theta}}{\pi_{\theta}}$$

这一步就是重要性采样



从而，我们说明了RL和OPD的目标函数在期望上具有一致性，因为优势函数对 θ 求导也为0，可以对RL的目标函数求导得出和左边一样形式的结果。从而我们证明了RL和OPD的优化目标在期望上具有一致性。

$$\begin{aligned} \nabla_{\theta} \mathcal{J}_{\text{RKL}}(\theta) &= \nabla_{\theta} \mathcal{J}_{\text{RKL}}^{(\text{sg})}(\theta) = \mathbb{E}_{q \sim \mathcal{Q}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \rho_{i,t}(\theta) R_{i,t}(\theta) \nabla \log \pi_{\theta}(o_t|q, o_{<t}) \right], \\ \mathcal{J}_{\text{RL}}(\theta) &= \mathbb{E}_{q \sim \mathcal{Q}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \rho_{i,t}(\theta) \hat{A}_{i,t} \right], \end{aligned} \quad (1)$$

那么问题接踵而至：方差上呢？可以见得， $\nabla_{\theta} R$ 在期望上为0，但可能提供非0方差。然而，在不改变期望的前提下，减掉一个期望为0、方差极大的干扰项，从而降低整体方差，反而是一件好事。

$$\nabla_{\theta} \mathcal{J}_{\text{RKL}} = \underbrace{\mathbb{E}[\rho_{i,t} R_{i,t} \nabla \log \pi_{\theta}]}_{\text{有效梯度项}} + \underbrace{\mathbb{E}[\rho_{i,t} \nabla_{\theta} R_{i,t}]}_{\text{噪声梯度项}}$$

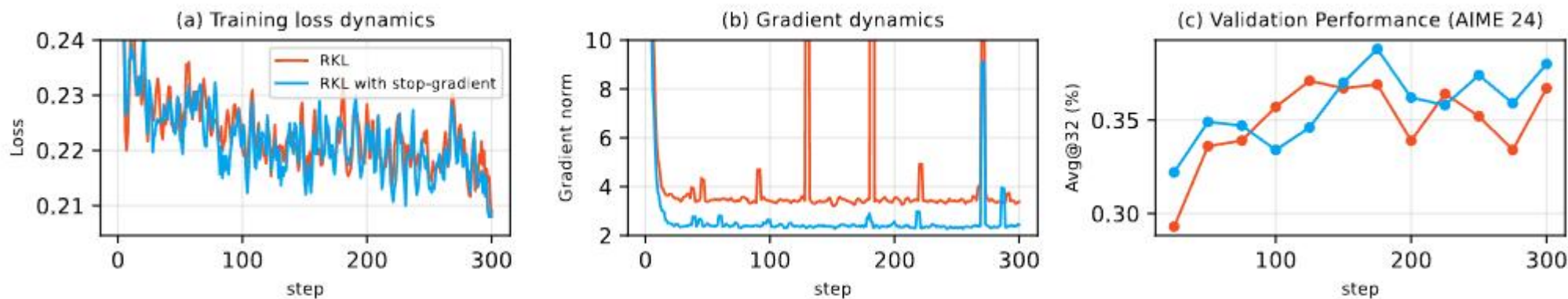


Figure 3. Comparison of training dynamics between vanilla RKL and RKL with stop-gradient. (a) Training loss dynamics exhibit similar trends, aligning with the theoretical equivalence in Remark 3.1. (b) The gradient norm is markedly lower and more stable when stop-gradient is applied, which (c) leads to superior validation performance. This confirms that treating the log-likelihood ratio as a fixed reward signal is beneficial for optimization stability.

图中也可以看出消除掉这一项的作用，从b中就可以看出去除这个高噪声项后梯度范数变化更稳了。



虽然停止梯度构造通过稳定梯度动态构建了一个强基线，但我们发现，同策略蒸馏仍然继承了RL所特有的根本性优化难题：

- 重尾负奖励引发的不稳定性：教师模型对词元的输出概率趋近于0时会出现极端负奖励，原因与前向KL的模式覆盖一样。
- 近零奖励导致的训练低效：对于绝大多数令牌，学生与教师的分布本就高度对齐，对应的对数似然比接近零，这类近乎消失的优势函数仅提供极微弱的学习信号，却仍占用批次内存与计算资源。
- 熵塌缩：熵值快速坍塌，导致模型过早收敛。

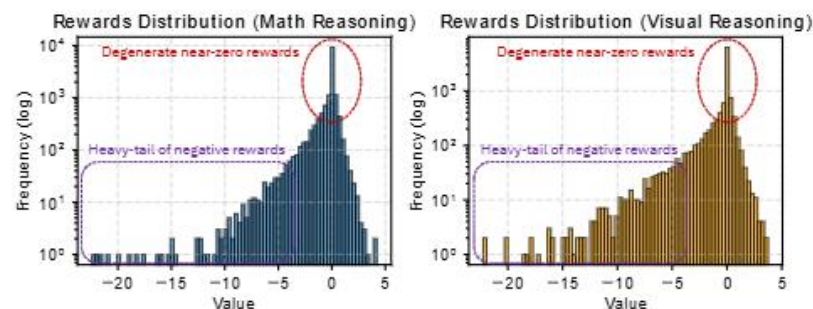


Figure 4. Log-scale histogram of token-level log-likelihood ratio rewards. The red dashed ellipse indicates degenerate near-zero rewards, while the purple dashed region highlights the heavy tail of negative rewards. These distributions pose challenges for RL-based on-policy distillation by causing gradient vanishing and training instability, respectively.

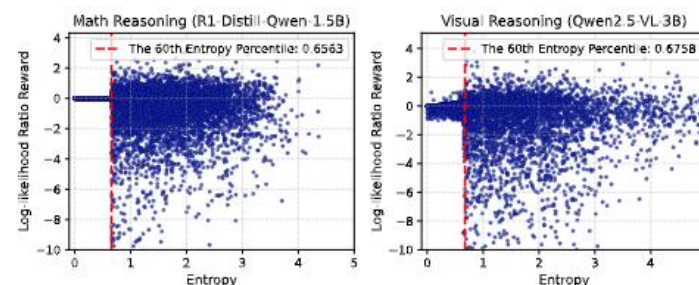


Figure 5. Correlation between token entropy and log-likelihood ratio rewards. Experimental results on math reasoning and visual reasoning benchmarks demonstrate that rewards in the bottom 60th entropy percentile are heavily concentrated around zero. This suggests that while teacher and student policies may diverge overall, they remain highly consistent on low-entropy, deterministic tokens, with significant deviations occurring primarily in high-entropy regimes.



奖励裁剪:

裁剪重要性采样比例是强化学习中的标准做法，但它仅能限制策略更新的幅度，无法保证学习信号本身的完整性。然而在OPD中，训练不稳定的主要根源是重尾奖励分布，因此我们需要对奖励进行裁剪。

先对奖励部分进行缩放，尝试寻求边界。由log为下凸函数可得：

$$(1 - \lambda) \cdot \log \pi_T(o_t|q, o_{<t}) + \lambda \cdot \log \pi_\theta(o_t|q, o_{<t}) \leq \log [(1 - \lambda) \cdot \pi_T(o_t|q, o_{<t}) + \lambda \cdot \pi_\theta(o_t|q, o_{<t})]. \quad (20)$$

进而：

$$R_{i,t}(\theta) = \log \frac{\pi_T(o_t|q, o_{<t})}{\pi_\theta(o_t|q, o_{<t})} \leq \frac{1}{1 - \lambda} \log \frac{(1 - \lambda)\pi_T(o_t|q, o_{<t}) + \lambda\pi_\theta(o_t|q, o_{<t})}{\pi_\theta(o_t|q, o_{<t})}.$$

左侧的原始奖励在教师模型概率趋近于0时会发散至负无穷，而右侧项则会收敛至有限常数 $\frac{\log \lambda}{1 - \lambda}$ 。

$$\hat{R}_{i,t}^\lambda(\theta) = \max \left(\text{sg}(R_{i,t}(\theta)), \frac{\log \lambda}{1 - \lambda} \right) \quad (7)$$



基于熵引导的词元级动态采样:

绝大多数词元的生成具有强确定性，教师与学生策略都会为其赋予几乎相同的高概率，进而导致梯度消失。与之相反，高熵词元往往对应推理的关键分支点能提供更具有价值的学习信号，这一点也有研究证明过了。

其实做法很简单，低熵处打上掩码，仅在高熵处计算梯度。

$$\mathcal{J}_{\text{Ent}}(\theta) = \mathbb{E}_{\mathbf{B} \sim \mathcal{Q}, q \sim \mathbf{B}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G \sum_{l=1}^{|o_i|} \mathbb{I}[H_l^i \geq \tau_\beta]} \sum_{i=1}^G \sum_{l=1}^{|o_i|} \rho_{i,l}(\theta) \hat{R}_{i,l}^\lambda(\theta) \mathbb{I}[H_l^i \geq \tau_\beta] \right]. \quad (8)$$

公式中H表示学生策略在词元上的熵， τ_β 表示为批次B内熵值前 β 分位的阈值。



多阶段训练:

通过动态掩码，设计了两阶段训练：前期为探索阶段，鼓励生成多样且合理的解（类似SFT）；后期为精炼阶段，聚焦并强化正确的推理路径（类似RL）。

$$\mathcal{J}_{\text{REOPOLD}}(\theta) = \mathbb{E}_{\mathcal{B} \sim \mathcal{Q}, q \sim \mathcal{B}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^G \sum_{t=1}^{|o_i|} M_{i,t}} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \rho_{i,t}(\theta) \hat{R}_{i,t}^\lambda(\theta) M_{i,t} \right].$$

训练初期，通过掩码过滤掉过度的惩罚： $M_{i,t} = \mathbb{I} \left[R_{i,t}(\theta) \geq \frac{\log \lambda}{1 - \lambda} \right]$ 。直观来看，消除过大的负梯度，模拟了SFT的训练动态：只强化正向行为，不对探索中的错误施加激进惩罚。

训练后期，通过过滤低熵： $M_{i,t} = \mathbb{I} [H_t^i \geq \tau_\beta]$ ，学习聚焦于高熵词元（即教师与学生策略差异最大的关键位置），助力策略精炼与收敛。通过在这些关键的不确定词元上允许负向反馈，能有效巩固模型学到的策略。

Algorithm 1 REOPOLD

- 1: **Input:** student policy π_θ , teacher policy π_T , query set $\mathcal{Q}$
- 2: **Hyperparameters:** total steps K , switch step T_{switch} , clipping coefficient λ , entropy percentile β , learning rate η
- 3: **Output:** trained student model π_θ
- 4: Initialize $\theta_{\text{old}} \leftarrow \theta$
- 5: **for** $k = 1$ to K **do**
- 6: Sample a batch of queries $\mathcal{B} \sim \mathcal{Q}$
- 7: Generate $\{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | q)$ for each $q \in \mathcal{B}$
- 8: Compute $R_{i,t}(\theta) \leftarrow \log \frac{\pi_T(o_{i,t}|q, o_{i,<t})}{\pi_\theta(o_{i,t}|q, o_{i,<t})}$
- 9: Clip $\hat{R}_{i,t}^\lambda(\theta) \leftarrow \max \left(\text{sg}(R_{i,t}(\theta)), \frac{\log \lambda}{1-\lambda} \right) \triangleright \text{Eq. (7)}$
- 10: **if** $k < T_{\text{switch}}$ **then**
- 11: *Phase I: Exploration (Reward-based Filtering)*
- 12: Set mask $M_{i,t}^{(k)} \leftarrow \mathbb{I} \left[R_{i,t}(\theta) \geq \frac{\log \lambda}{1-\lambda} \right] \triangleright \text{Eq. (9)}$
- 13: **else**
- 14: *Phase II: Refinement (Entropy-Guided Sampling)*
- 15: Compute entropy H_t^i for each token $o_{i,t}$ in batch
- 16: Compute τ_β as top β -percentile of H_t^i
- 17: Set mask $M_{i,t}^{(k)} \leftarrow \mathbb{I} [H_t^i \geq \tau_\beta] \triangleright \text{Eq. (10)}$
- 18: **end if**
- 19: Compute gradients $\nabla_\theta \mathcal{J}_{\text{REOPOLD}}$ using Eq. (4)
- 20: Update parameters $\theta \leftarrow \theta + \eta \nabla_\theta \mathcal{J}_{\text{REOPOLD}}$
- 21: Update old policy parameters $\theta_{\text{old}} \leftarrow \theta$
- 22: **end for**

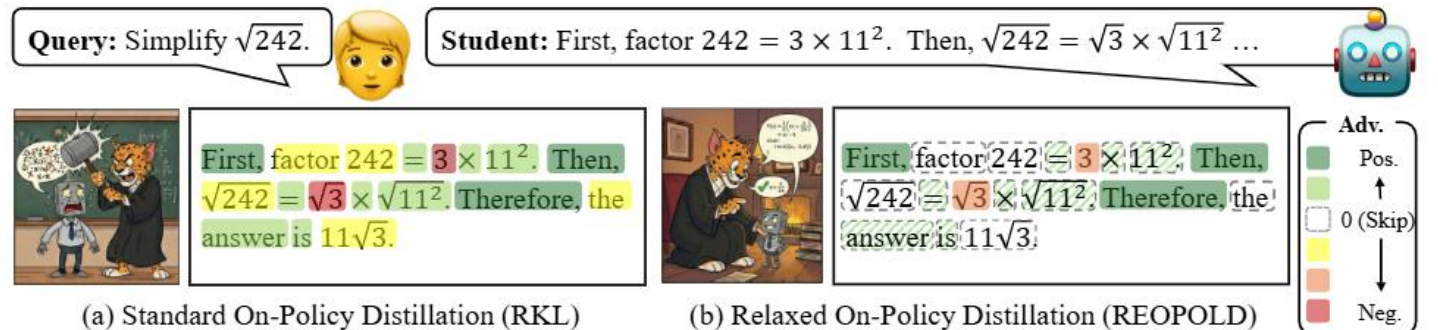


Figure 2. **Illustration of REOPOLD.** While standard on-policy distillation (a) often introduces instability and inefficiency by forcing the student to mimic the teacher excessively, our proposed REOPOLD (b) fosters a more stable and effective learning environment. By establishing a formal connection between distillation and RL via a stop-gradient operation (Section 3.1), REOPOLD utilizes teacher signals temperately (Section 4.1) and selectively (Section 4.2). As depicted, this approach filters out potentially harmful signals, preventing the student from deviating excessively from its original distribution.

$$\mathcal{J}_{\text{REOPOLD}}(\theta) = \mathbb{E}_{\mathcal{B} \sim \mathcal{Q}, q \sim \mathcal{B}, \{o_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot | q)} \left[\frac{1}{\sum_{i=1}^G \sum_{t=1}^{|o_i|} M_{i,t}^{(k)}} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \rho_{i,t}(\theta) \hat{R}_{i,t}^\lambda(\theta) M_{i,t}^{(k)} \right], \quad (4)$$

where

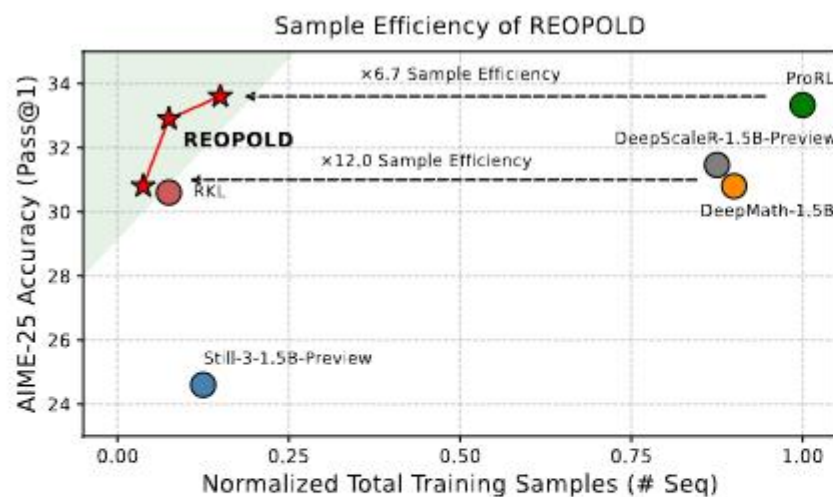
$$\hat{R}_{i,t}^\lambda(\theta) = \max \left(\text{sg}(R_{i,t}(\theta)), \frac{\log \lambda}{1-\lambda} \right), \quad (5)$$

$$M_{i,t}^{(k)} = \begin{cases} \mathbb{I} \left[R_{i,t}(\theta) \geq \frac{\log \lambda}{1-\lambda} \right] & \text{if } k < T_{\text{switch}} \\ \mathbb{I} [H_t^i \geq \tau_\beta] & \text{if } k \geq T_{\text{switch}} \end{cases}, \quad (6)$$

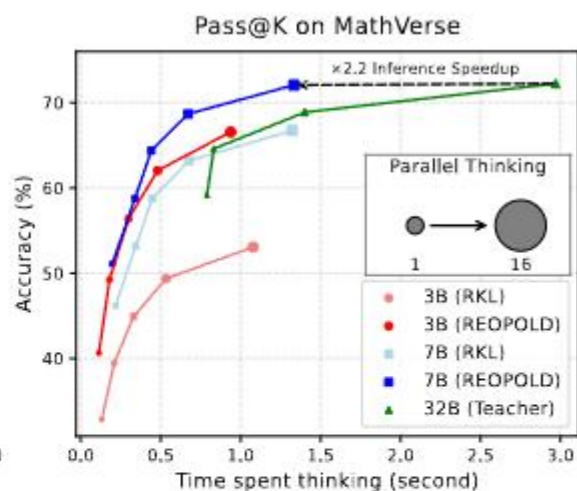
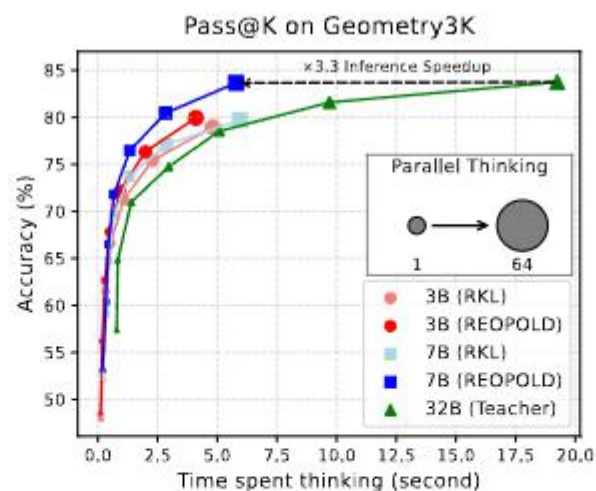


实验设置:

- 学生模型: 文本: DeepSeek-R1-Distill-Qwen-1.5B 和 7B; 视觉推理: Qwen2.5-VL-3/7B-Instruct
- 教师模型: 文本: SkyWork-OR1 (-Math)-7B 与 SkyWork-OR1-32B-Preview; 视觉推理: 经过 NoisyRollout训练的 Qwen2.5-VL-32B-Instruct
- 训练数据: 文本: LUFFY, 包含 45k 条提示词的数据集; 视觉: Geometry3K



(a) REOPOLD on DeepSeek-R1-Distill-Qwen-1.5B



(b) Test-time scaling of REOPOLD

Figure 1. Performance of REOPOLD. (a) **Sample Efficiency:** REOPOLD achieves a state-of-the-art trade-off between accuracy and sample efficiency on the AIME-25 benchmark. Detailed explanation can be found in Section 5.1. (b) **Test-time Scaling:** On visual reasoning tasks, REOPOLD demonstrates superior test-time scaling capabilities compared to the vanilla RKL baseline. Notably, it allows smaller models to approach the performance of the 32B teacher. Detailed explanation can be found in Section 5.2.



Table 1. Performance comparison of REOPOLD and distillation baselines (SFT and RKL) for on mathematical reasoning benchmarks across different teacher models. Accuracy (%) is reported for all benchmarks. The best result in each column is shown in **bold**, and the second-best is underlined. $\Delta\%$ indicates relative improvement of REOPOLD compared to RKL. The RL-based approach is colored in gray.

Model	AIME-24	AIME-25	AMC-23	MATH-500	Minerva Math	Olympiad Bench	AVG.
SkyWork-OR1-Math-7B (He et al., 2025a) $\rightarrow$ DeepSeek-R1-Distill-Qwen-1.5B (Guo et al., 2025)							
SkyWork-OR1-Math-7B	69.8	52.3	94.1	95.8	49.3	73.5	72.5
R1-Distill-Qwen-1.5B	28.6	22.7	62.6	82.9	26.4	43.6	44.4
+ GRPO	31.8	23.7	62.0	85.4	33.8	49.8	47.8
+ SFT	33.5	24.6	76.1	86.6	<u>36.4</u>	55.6	51.5
+ RKL	<u>37.1</u>	<u>30.6</u>	<u>80.2</u>	<u>88.0</u>	34.6	<u>56.0</u>	<u>54.4</u>
+ REOPOLD	41.6	32.6	83.0	89.2	38.6	57.3	57.1
$\Delta\%$	+12.13%	+6.54%	+3.49%	+1.36%	+11.56%	+2.32%	+4.96%
SkyWork-OR1-7B (He et al., 2025a) $\rightarrow$ DeepSeek-R1-Distill-Qwen-1.5B (Guo et al., 2025)							
SkyWork-OR1-7B	65.3	49.7	91.8	95.4	47.1	72.5	70.3
+ RKL	31.9	23.9	63.0	81.6	30.9	47.9	46.5
+ REOPOLD	36.2	26.7	78.1	84.2	30.2	53.3	51.5
$\Delta\%$	+13.48%	+11.71%	+23.97%	+3.19%	-2.27%	+11.27%	+10.75%



Table 2. Performance comparison of vision-language models on a visual reasoning and perception benchmarks. Accuracy (%) is reported for all benchmarks. The best result in each column is shown in **bold**, and the second-best is underlined. [†] denotes the results from Liu et al. (2025b) as original model is not available. $\Delta\%$ indicates relative improvement of REOPOLD compared to RKL. The RL-based approaches are colored in gray.

Model	Geo3K	MathVerse	MathVision	MathVista	WeMath	Hallusion	AVG.
Qwen2.5-VL-32B-Instruct + NoisyRollout	56.74	58.86	39.82	78.30	75.51	72.45	63.61
Qwen2.5-VL-32B-Instruct + NoisyRollout (Liu et al., 2025b) $\rightarrow$ Qwen2.5-VL-3B-Instruct							
Qwen2.5-VL-3B-Instruct	26.46	35.58	22.83	59.40	53.41	61.51	43.20
+ PAPO (Wang et al., 2025b)	32.95	<u>40.65</u>	24.16	<u>65.10</u>	63.62	<u>61.62</u>	<u>48.02</u>
+ RKL	<u>48.09</u>	<u>33.25</u>	<u>24.78</u>	<u>62.50</u>	<u>64.48</u>	<u>60.83</u>	<u>48.99</u>
+ REOPOLD	50.58	46.40	26.39	61.50	64.60	63.62	52.18
$\Delta\%$	+2.06%	+26.44%	+3.39%	-1.60%	+0.19%	+4.59%	+4.27%
Qwen2.5-VL-32B-Instruct + NoisyRollout (Liu et al., 2025b) $\rightarrow$ Qwen2.5-VL-7B-Instruct							
Qwen2.5-VL-7B-Instruct	39.77	45.72	25.05	67.80	64.77	65.62	51.46
+ GRPO [†] (Shao et al., 2024)	51.4	50.8	27.3	70.5	67.4	69.8	56.20
+ NoisyRollout (Liu et al., 2025b)	50.08	<u>53.14</u>	26.64	<u>72.00</u>	<u>70.57</u>	<u>70.66</u>	<u>57.18</u>
+ RKL	<u>51.75</u>	47.71	<u>28.79</u>	<u>71.27</u>	<u>70.06</u>	69.51	<u>56.51</u>
+ REOPOLD	53.58	<u>51.43</u>	29.21	72.40	69.77	<u>70.14</u>	57.76
$\Delta\%$	+3.54%	+7.80%	+1.46%	+1.59%	-0.41%	+0.91%	+2.21%

Table 3. Performance comparison of Qwen2.5-VL-3B-Instruct trained with various teacher models on a visual reasoning and perception benchmarks. Accuracy (%) is reported for all benchmarks.

	RKL	REOPOLD	$\Delta\%$
Qwen2.5-VL-7B-Instruct + NoisyRollout			
Geo3K	49.75	51.41	+1.66
MathVerse	41.66	44.27	+2.61
MathVision	23.33	24.01	+0.68
MathVista	62.20	63.10	+0.90
WeMath	64.83	65.66	+0.83
Hallusion	61.30	62.67	+1.37
AVG.	50.51	51.85	+1.34
Qwen2.5-VL-32B-Instruct + NoisyRollout [†]			
Geo3K	43.93	45.76	+1.83
MathVerse	41.99	43.22	+1.23
MathVision	25.13	25.79	+0.66
MathVista	63.60	64.20	+0.60
WeMath	64.02	64.66	+0.64
Hallusion	63.72	64.35	+0.63
AVG.	50.40	51.33	+0.93

Table 4. Performance comparison of Qwen2.5-VL-3B-Instruct trained for 60 and 300 training steps on a visual reasoning and perception benchmarks. Accuracy (%) is reported for all benchmarks.

	RKL	REOPOLD	$\Delta\%$
60 Training Steps (3B)			
Geo3K	48.09	50.58	+2.06
MathVerse	33.25	46.40	+26.4
MathVision	24.78	26.39	+3.39
MathVista	62.50	61.50	-1.60
WeMath	64.48	64.60	+0.19
Hallusion	60.83	63.62	+4.59
AVG.	48.99	51.08	+4.27
300 Training Steps (3B)			
Geo3K	49.08	51.08	+2.00
MathVerse	46.60	47.79	+1.19
MathVision	26.16	27.44	+1.28
MathVista	67.20	66.30	-0.90
WeMath	66.43	67.18	+0.75
Hallusion	65.19	66.35	+1.16
AVG.	53.44	54.36	+0.92

Table 5. Performance comparison of Qwen2.5-VL-7B-Instruct trained for 60 and 300 training steps on a visual reasoning and perception benchmarks. Accuracy (%) is reported for all benchmarks.

	RKL	REOPOLD	$\Delta\%$
60 Training Steps (7B)			
Geo3K	51.75	53.58	+3.54
MathVerse	47.71	51.43	+7.80
MathVision	28.79	29.21	+1.46
MathVista	71.27	72.40	+1.59
WeMath	70.06	69.77	-0.41
Hallusion	69.51	70.14	+0.91
AVG.	56.51	57.76	+2.21
300 Training Steps (7B)			
Geo3K	49.42	53.58	+4.16
MathVerse	50.72	51.97	+1.25
MathVision	29.74	31.12	+1.38
MathVista	71.20	73.60	+2.40
WeMath	69.43	71.84	+2.41
Hallusion	69.72	70.87	+1.15
AVG.	56.71	58.83	+2.12

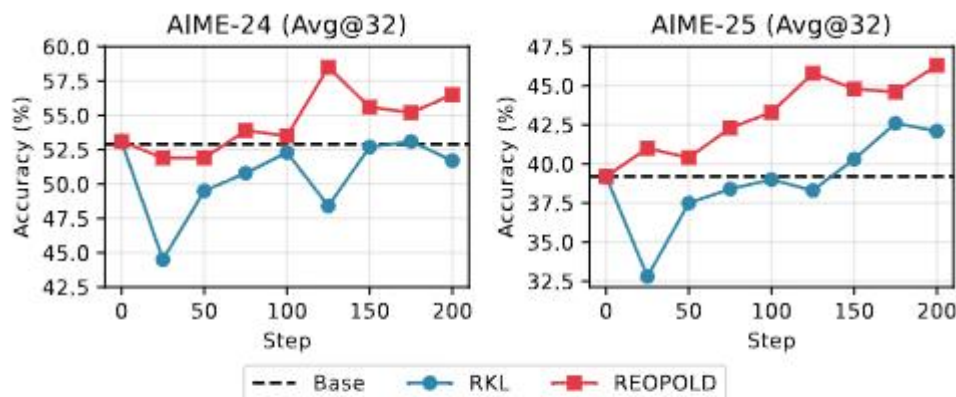


Figure 7. Validation accuracy on AIME with a 7B student policy. With a 7B student, vanilla RKL becomes unstable, whereas REOPOLD enables stable and consistent improvement.

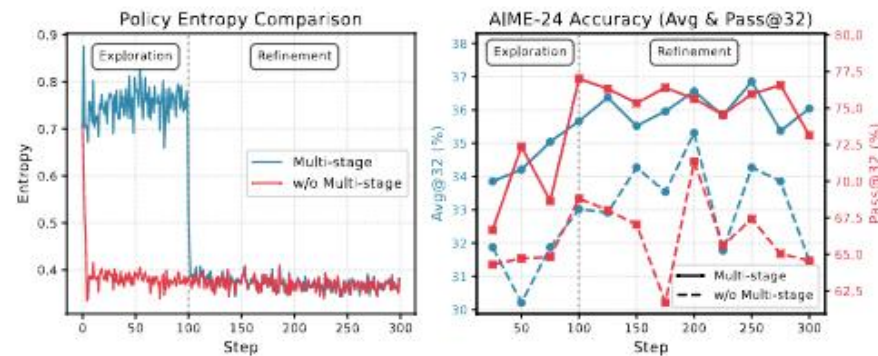


Figure 6. Analysis of exploration-to-refinement multi-stage training. (Left) The proposed multi-stage approach prevents entropy collapse and preserves diverse modes during the initial exploration phase by masking strongly negative rewards, whereas the baseline suffers from rapid entropy collapse. (Right) The multi-stage strategy (solid lines) achieves superior final performance to the baseline (dashed lines) in both quality (i.e., Avg@32) and diversity (i.e., Pass@32).

Table 6. Performance comparison of Qwen2.5-VL-3B-Instruct trained with various teacher models on a visual reasoning and perception benchmarks. Accuracy (%) is reported for all benchmarks.

	(1)	(2)	(3)	(4)	Geo3K	Verse	Vision	AVG.
Qwen2.5-VL-3B-Instruct								
RKL					48.09	33.25	24.78	35.37
	✓				48.42	43.55	26.02	39.33
	✓	✓			48.59	45.53	25.07	39.73
	✓	✓	✓		<u>50.08</u>	<u>45.94</u>	<u>26.42</u>	<u>40.81</u>
REOPOLD	✓	✓	✓	✓	50.58	46.40	<u>26.39</u>	41.12
Qwen2.5-VL-7B-Instruct								
RKL					51.75	47.71	28.79	42.75
	✓				52.08	49.52	28.12	43.24
	✓	✓			51.75	51.45	29.05	44.08
	✓	✓	✓		<u>52.75</u>	51.42	<u>29.10</u>	<u>44.42</u>
REOPOLD	✓	✓	✓	✓	53.58	<u>51.43</u>	29.21	44.74

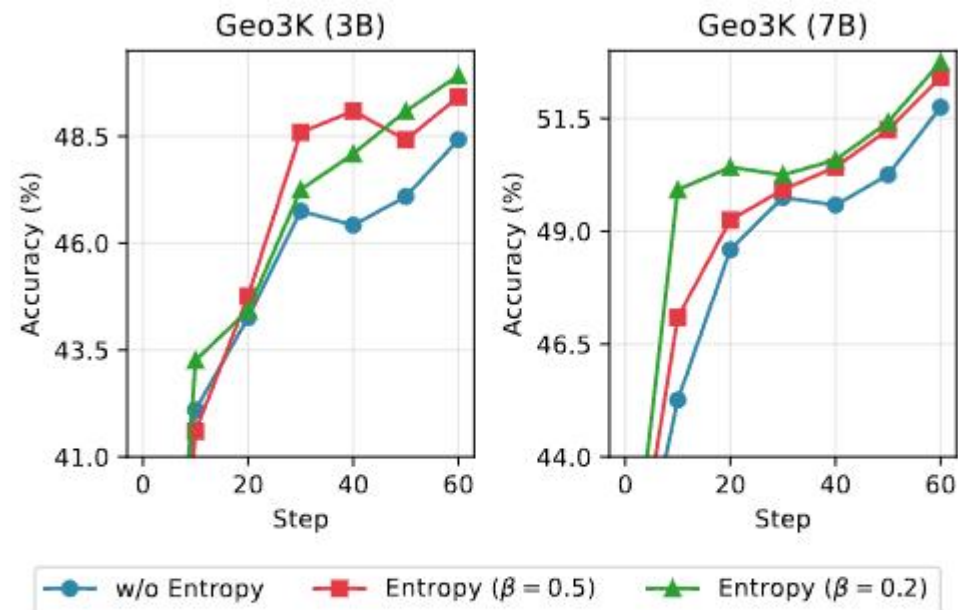


Figure 8. Comparison among REOPOLD using different range of tokens. Our token selection accelerates convergence and leads to superior final accuracy.

- 正常来说OPD之前需要先进行SFT或RL，但这篇的多阶段训练设置可以省略这一点
- 本篇数学推导很精彩，虽然后续方法论感觉像缝合怪，但总体感觉还算一篇佳作
- OPD过程中，如果学生模型生成的轨迹严重偏离教师模型的分布，教师模型是否还能提供有效的指导？
- 感觉OPD很实用于刷榜，是否具有代替强化学习的潜力，用更少资源训练出效果更好的模型？

1. [A Survey of On-Policy Distillation for Large Language Models](#)
2. [On-Policy Distillation](#)
3. [On Policy distillation of language models: Learning from self-generated mistakes](#)
4. [Scaling Reasoning Efficiently via Relaxed On-Policy Distillation](#)